Manufactured Matches: Generative Face Restoration Inflates False Alarms in Open-Set Low-Resolution Recognition

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***Abstract*—Operators of watchlist face recognition routinely receive low-resolution probes, from distant cameras or compressed video, and a common response is to enhance the probe with a super-resolution or face-restoration model before matching. We measure what this does to open-set identification, where a probe may belong to no enrolled person, using frozen public face encoders and public data. Three results follow. First, no post-hoc processing of a single degraded probe recovers the lost identity: pooling many frames (mean or quality-weighted) saturates at a resolution-set ceiling, and single-image restoration does not cross it. Second, blind restoration actively lowers true identification at every face size from 8 to 48 pixels. Third, and most consequential, restoration guided by a gallery candidate transfers that candidate's identity onto the probe, so that a non-enrolled stranger is accepted as the candidate 6 to 14 times above the fixed 1% false-alarm baseline, with no gain in true detection. The hazard is worst at the lowest resolutions and resists lightweight mitigation, because a low-resolution probe is itself identity-underdetermined, making a manufactured match indistinguishable from a real one. The single lever that crosses the resolution ceiling is genuine multi-frame super-resolution that fuses real sub-pixel samples, at a measured best-case exchange rate of roughly 8 to 32 frames per resolution step. We give a practitioner rule: do not restore a single low-resolution face before open-set search with a frozen matcher.**

***Index Terms*—Open-set face recognition, watchlist identification, low-resolution face recognition, face super-resolution, face restoration, generative models, biometric security.**

I. Introduction

A watchlist system answers a harder question than verification: given a probe face, decide whether it belongs to one of the enrolled identities and, if so, which, while rejecting the far larger population of non-enrolled people. The decisive metric is the true positive identification rate at a fixed false positive identification rate (TPIR at a given FPIR). Probes from surveillance are often small, a face a few dozen pixels across, which already degrades one-to-many identification [1], and a natural engineering reflex is to enhance the probe with a learned super-resolution or face-restoration model and then match. We ask whether this helps or harms open-set identification, and we find that it harms it in two different ways, one of which is a security hazard rather than a mere loss of accuracy.

Our contributions are as follows.

- We show that **reference-guided restoration manufactures matches**: restoring a probe toward a gallery candidate transfers that candidate's identity, so strangers are accepted as the candidate 6 to 14 times above baseline at a fixed FPIR, with no detection gain. We show the hazard resists two lightweight defenses and explain why.
- We establish a **resolution ceiling** that explains the hazard: pooling frames saturates below the next resolution step, blind single-image restoration only hurts, and gallery-guided restoration adds no true detection; only genuine multi-frame super-resolution crosses, at a measured frames-for-pixels exchange rate.
- We give a **practitioner rule** and an operating envelope: do not restore a single low-resolution face before open-set search; below about 16 pixels the only safe policy is to abstain.

II. Setup

We build an open-set watchlist on YouTube Faces (1,595 identities) [2] and the ChokePoint surveillance set [3], with two frozen public encoders, ArcFace (InsightFace `buffalo_l`, R50) [4] and MagFace (R100) [12], the latter for a cross-encoder robustness check. Probes are degraded to a controlled ladder (the face resized to 8, 12, 16, 24 or 48 pixels and back to the encoder input) so the only variable is resolution. Each enrolled identity contributes a high-resolution gallery photo; known probes come from a different video than enrollment; unknown probes come from held-out identities. We report rank-1, AUROC between known and unknown probes, and TPIR at fixed FPIR. Cascade thresholds are chosen by two-fold cross-fitting over identities, and confidence intervals are identity-clustered bootstraps. Every comparison within a figure is computed in one pass on one split so numbers are directly comparable.

III. The resolution ceiling: frames do not buy pixels

On 1,595 identities (800 enrolled, 795 unknown), mean-pooling the per-frame embeddings of a track raises accuracy with more frames but plateaus by about sixteen frames, far below a single frame one resolution step higher (Table I;

TABLE I

POOLING AT 8 PX NEVER REACHES A SINGLE 12 PX FRAME (YOUTUBE FACES, ARCFACE). REPORTED AT 10% FPIR; AT 8 PX THE 1% POINT USED IN TABLES III AND IV IS NEAR THE DETECTION FLOOR. BEST-N IS THE BEST OVER N UP TO 32.

metric	best pooling @ 8 px	single frame @ 12 px
rank-1	0.105	**0.436**
AUROC (known vs unknown)	0.535	**0.685**
TPIR @ 10% FPIR	0.041	**0.282**

TABLE II

RANK-1 BY FRAMES POOLED N AND PROBE RESOLUTION (YOUTUBE FACES, ARCFACE), THE PER-(N, RESOLUTION) BREAKDOWN OF TABLE I. EACH COLUMN PLATEAUS WITH N, BELOW THE SINGLE-FRAME VALUE OF THE COLUMN TO ITS LEFT.

N \ resolution	HR	16 px	12 px	8 px
1	0.926	0.739	0.436	0.061
2	0.932	0.790	0.505	0.092
4	0.935	0.814	0.532	0.100
8	0.938	0.816	0.560	0.102
16	0.936	0.823	0.565	0.105
32	0.930	0.813	0.553	0.094

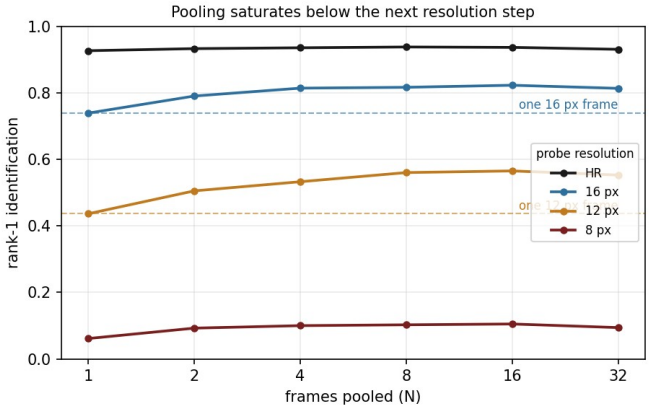


Fig. 1. Pooling saturates below the next resolution step. Rank-1 against frames pooled for each probe resolution; dashed lines mark a single frame at the next-higher resolution, which pooling the level below never reaches.

Table II and Fig. 1 break this out by frame count and resolution). Quality-weighted pooling, weighting frames by the ArcFace magnitude, gives the same answer.

The same saturation holds on ChokePoint surveillance tracks. The one aggregation that does cross the ceiling is genuine multi-frame super-resolution: fusing real sub-pixel-shifted low-resolution samples in pixel space (best case, with known shifts) raises 8 px identification past a single 12 px frame, and 12 px past a single 16 px frame, at an exchange rate of roughly 8 to 32 frames per step (ChokePoint, 15 enrolled and 14 unknown identities, gallery padded with 400 external distractors). Averaging embeddings cannot do this because degradation is a shared bias, not independent variance; only new sub-pixel information crosses.

A modern learned fuser shows the same split on real tracks. Running BasicVSR++ [5], a video super-resolution network, on 29 consecutive ChokePoint tracks (15 known, 14 unknown) raises the mated similarity of the restored probe to its true identity monotonically with the number of frames fused (at 8 px, from 0.37 for one frame to 0.50 for sixteen), while the same network given a single frame lowers it below the raw low-resolution frame (0.41 down to 0.37). The dividing line is the source of the added detail: fusing several frames adds real sub-pixel information and helps, consistent with multi-frame face super-resolution raising identification [6], whereas single-image restoration invents detail and hurts.

TABLE III

CHANGE IN TPIR @ 1% FPIR FROM RESTORING EVERY PROBE, VERSUS THE LOW-RESOLUTION BASELINE (YOUTUBE FACES, 800 ENROLLED AND 800 UNKNOWN PROBES FROM 400 + 400 IDENTITIES). NEGATIVE IS WORSE. EVERY CELL IS A LOSS.

face size	LR baseline	GFPGAN	CodeFormer	Real-ESRGAN
16 px	0.522	−0.371	−0.266	−0.501
24 px	0.787	−0.240	−0.177	−0.494
48 px	0.868	−0.080	−0.075	−0.045

IV. BLIND RESTORATION LOWERS IDENTIFICATION

We next apply three off-the-shelf restorers (GFPGAN [7], CodeFormer [8], Real-ESRGAN [9]) to the probe and re-match. Across face sizes from 8 to 48 pixels, restoring every probe lowers TPIR at 1% FPIR relative to matching the low-resolution probe directly (Table III shows 16 to 48 px; the loss is larger still at 8 and 12 px); the loss shrinks as faces grow but never becomes a gain, and a cross-fitted confidence-gated cascade selects "do not restore." Restoration does not create false alarms here (the false-alarm rate stays near 1%); it destroys true hits by replacing the real face with a sharp, confident, different one.

This holds on genuinely native low resolution, not only synthetic degradation. On TinyFace [10], a surveillance set of native face crops around 20 pixels, matching native probes against a native gallery gives rank-1 0.660 and TPIR at 1% FPIR 0.193; restoring the probes with GFPGAN lowers these to rank-1 0.571 and TPIR 0.013. Restoring the incoming probe moves it off the native gallery distribution, so the watchlist operating point falls sharply.

V. REFERENCE-GUIDED RESTORATION MANUFACTURES MATCHES

The sharper hazard appears when restoration is guided by a reference, as a watchlist naturally invites: restore the probe toward its top gallery candidate, then re-match. We use a reference-based latent-diffusion restorer (ReF-LDM [11]) guided by a photo of the candidate from the enrollment video. Guidance transfers the candidate's identity onto the probe (Fig. 2). At a threshold calibrated to 1% FPIR on the low-resolution probe, a non-enrolled stranger is accepted as its guiding candidate far above the baseline rate (Table IV).

Guidance buys no true detection to offset this: held at 1% FPIR, guided restoration matches the low-resolution baseline

TABLE IV
RATE AT WHICH A NON-ENROLLED STRANGER IS ACCEPTED AS ITS GUIDING CANDIDATE, AT A FIXED 1% FPIR OPERATING POINT (YOUTUBE FACES, 800 STRANGERS). GUIDED RESTORATION IS 6 TO 14 TIMES THE 1% TARGET BASELINE.

face size	low-res probe	blind restoration	guided restoration
8 px	0.011	0.001	**0.141**
12 px	0.011	0.001	**0.104**
16 px	0.011	0.001	**0.064**

at 8 px and falls below it at 12 and 16 px. A wrong guide pulls the restoration toward the wrong identity about 60% as strongly as the correct guide pulls toward the truth, and equally for enrolled and non-enrolled probes, so the restorer confirms whatever candidate it is given.

The hazard is robust in two ways. It persists when the guiding reference is drawn from a separate recording session rather than the enrollment video, at 0.052, 0.078 and 0.020 for 8, 12 and 16 pixel faces, still several times the baseline though milder, so its strength scales with how much the reference resembles the gallery photo. And it transfers across encoders: a second matcher (MagFace [12], a different architecture and loss) reading the same restorations accepts the manufactured identity at matching rates, so the injected identity is not an artifact of one encoder.

VI. THE HAZARD RESISTS LIGHTWEIGHT MITIGATION

Two natural gates fail where the hazard is worst. A guidance-boost gate (reject when the guided score exceeds the pre-restoration score by a margin) catches about 90% of manufactured matches but discards 34 to 69% of true hits as resolution drops, because the two populations overlap below 16 px and separate only at 16 px. A guide-promiscuity gate (reject probes that also accept a second random guide) does not separate them at all. The reason is structural: at low resolution a genuine probe is itself compatible with many identities, so a manufactured match is statistically indistinguishable from a real one. The only robust policy below about 16 pixels is to abstain.

VII. MECHANISM

All of the above follow from one property of the embedding space, which we call the unknownness barrier: as resolution drops, the embeddings of degraded enrolled probes and of genuine unknowns migrate into the same region, so the known-versus-unknown AUROC falls toward chance. The identity that resolution removes is not recoverable from the pixels of one frame: pooling cannot invent it, single-image restoration can only hallucinate a plausible substitute, and a reference supplies an identity from outside the probe. The exception, genuine multi-frame super-resolution, works precisely because it injects real sub-pixel information that one frame lacked.

VIII. SCOPE

We report two frozen public encoders, ArcFace (R50) and MagFace (R100, a different architecture and loss); the resolution ceiling of Section III and the manufactured-matches hazard of Section V both hold for the two. The hazard also holds when the guiding reference is drawn from a separate recording session rather than the enrollment video, though it is milder, so its strength scales with how much the reference resembles the gallery photo. We use synthetic resolution control across 8 to 48 pixels and confirm the blind-restoration result on native low resolution (TinyFace). The reference-guided hazard is evaluated on the synthetic ladder, not on native sub-20-pixel crops, where the facial landmarks that the 512-input restorer requires are not reliably recoverable; the blind restorers, which detect and align internally, carry the native test. Each reference-guided cell is a single run with restoration noise held fixed, over 800 unknown probes that resolve the operating point to about 0.1% FPIR. The pixel-space multi-frame super-resolution result is a best case with known sub-pixel shifts; a learned video super-resolution model (BasicVSR++) on real consecutive tracks confirms the direction below that bound, raising mated similarity monotonically with the number of frames fused.

IX. RELATED WORK

Super-resolution and face restoration have been reported to improve face recognition (surveyed by Jiang and colleagues [13]), and we do not dispute those results in their setting. The gains appear in 1:1 verification or closed-set identification, when the recognizer is trained jointly with or on the restored images (for example cascaded super-resolution with identity priors [14]), or when the degradation is synthetic down-sampling matched to the restorer's training distribution. Even with a frozen matcher, blind restoration can raise cross-quality verification accuracy, conditioned on the restoration preserving identity [15].

A separate line reports the opposite in the regime we study: a frozen off-the-shelf matcher, native low resolution, and restoration applied to the probe alone. The surveillance challenge that defines open-set watchlist evaluation [16] finds pre-GAN super-resolution neutral to slightly negative, and cross-resolution recognition degrades when probe and gallery resolutions mismatch (Massoli and colleagues [17]); forensic studies [18] find modern restoration raises the log-likelihood-ratio cost and that AI-driven restoration does not improve, and can harm, recognition by injecting invented detail, as generative upsampling is known to hallucinate identity-changing detail (PULSE [19]); and a study on native low resolution finds a learned super-resolution front-end does not beat feeding the aligned low-resolution probe directly (Luevano and colleagues [20]).

Reference-based restorers are known to over-rely on the guiding image: recent work reports that a mismatched reference makes the restorer copy the reference's appearance (IConFace [21]), and that reference dominance grows with degradation severity (FaceMe [22]). That work treats reference overuse as an image-fidelity artifact to be reduced. Our contribution is to measure its open-set recognition consequence directly, the true and false identification rates on enrolled and

non-enrolled probes under modern blind and reference-based generative restorers, which this literature does not report. The central result is the manufactured-match hazard: a reference-guided restorer transfers the candidate's identity and inflates the false-identification rate on strangers, a failure mode that cannot appear in verification or closed-set evaluation and is invisible to accuracy or rank-1.

X. Conclusion

Enhancing a low-resolution probe before open-set search with a frozen matcher does not recover identity, and when the enhancement is guided by a gallery candidate it manufactures matches, transferring the candidate's identity onto a stranger and inflating false identifications at a fixed operating point. The practitioner rule is to match the low-resolution probe directly and to abstain below about sixteen pixels, where a genuine probe is itself identity-underdetermined. The one enhancement that adds identity rather than inventing or importing it is genuine multi-frame super-resolution, which fuses real sub-pixel detail from several frames.

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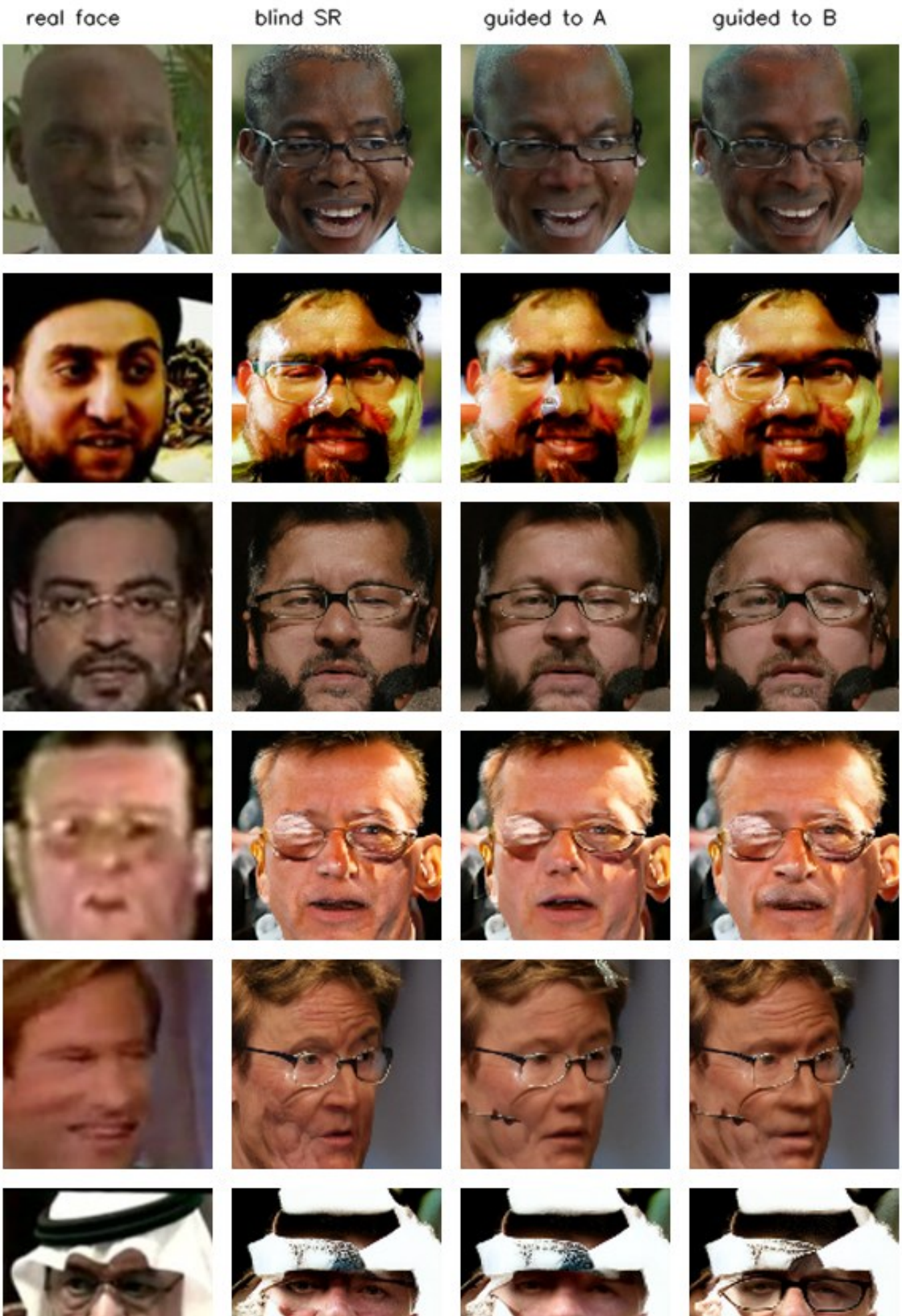
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real face
blind SR
guided to A
guided to B